

# Sollu Complete Feature Overview

Implemented features through 28 September 2026

Tamil-first communication support with English output and Tamil, English and Tanglish lookup. The local app combines everyday communication, caregiver support, rehabilitation practice and clinician review for aphasia, dysarthria, ALS, Parkinson's, laryngectomy and other communication needs.

## Communication and vocabulary

Speak, Type, Topics and Camera inputs; partner questions; body-part and side-specific comfort phrases; a searchable catalogue of 124 controlled meanings; saved favourites, recent messages and draft recovery. Personal words support aliases, descriptions, pinning, hiding, editing and deletion, with patient approval before everyday use and renewed approval after edits or imports. Fragment repair handles stutters, repeated words, syllable splits and uniquely supported prefixes while preserving the original input and showing the proposed reading.

## Personal communication tools

Photo scenes with selectable hotspots and equivalent large message lists; prepared conversation stories and scripts; a drawing and writing board; printable communication passport; request and refusal sentence builder; personal contacts and listener selection. Persistent Help, Fix, Pause and Stop controls support repair, editing, clarification and partner-understanding checks. Help has prepared wording and local sound or voice fallbacks, with honest delivery status and an optional manually sent SMS draft.

## AI sentence choices

Free catalogue suggestions and optional local Ollama, plus OpenAI, Claude, Gemini and Groq adapters. API responses offer up to three distinct supported choices, with Show more options, None of these, and separate AI-draft or prepared-phrase labels. Grounding checks preserve refusals, body side, numbers, language and stated details; duplicates and rejected meanings are filtered. Uncertain input can request clarification or yield no suggestion. Only an explicit choice speaks; previews are optional, and changed input cancels stale suggestions.

## Context and reviewed personalization

Time of day and morning-to-night periods, manual place without GPS, listener, partner question, personal preferences and optionally confirmed recent turns inform suggestions. Caregiver-reviewed routines include weekdays and time/place windows, edit/delete/undo controls and visible context clues. Repeated confirmed everyday logs can propose routines for review; therapy repetitions are excluded. Approved heard-to-intended corrections can be scoped by language, place and listener. Explicit words override routine hints; routines never prove that an event happened.

## Speech input and personal voice

Local versus Google/browser-online recognition is selectable for communication and practice. Local recognition requires browser support and an installed language pack; online use needs explicit permission, with no silent cloud fallback. Type, Topics and manual transcript review remain available. Device text-to-speech offers language, voice and speed controls. Voice Studio records, previews, approves, replays and deletes consented exact-phrase recordings. Camera object recognition runs locally after its model download; local-only protection blocks external model downloads.

## Caregiver companion and accessible settings

QR/private-link pairing connects a separate caregiver device through encrypted messages, a durable retry outbox and delivery/acknowledgement states. Remote questions, recent-word feed, Help cancellation, I am coming, and optional alarm/vibration require an enabled, open companion page. Settings cover large controls, text size, contrast, preferred hand, quiet screen, reduced motion, tap filtering, listening pauses, speech rate, first input, glosses, one-to-three choices, two-step confirmation and brief/natural/polite sentence styles. A local PIN limits accidental caregiver-setting changes.

# Rehabilitation and Clinician Support

## Rehabilitation practice

A dedicated Rehabilitation tab links overview, progress and practice. Editable goals, instructions, weekly targets, short-session plans and rest reminders support natural speech, AAC, mixed communication, electrolarynx, TEP and oesophageal speech. Curated or custom Tamil/English words, sentences, scripts, AAC tasks, confirmed messages and marked words become practice targets. Skip, rest and stop remain available. Optional audio/video has per-attempt consent, preview, discard and save, with two-minute/20 MB capture limits. Attempts retain their target, method and context, reviewed transcript, effort, fatigue and person/partner feedback.

## Progress and missed words

Seven- and 28-day views provide language/method filters, weekly-goal progress, active-day calendar, recent attempts and links from missed words into practice. Dashboards show reviewed transcript text matching, word matches, omissions, substitutions and extra words, plus response-time summaries, effort/fatigue, understanding feedback, AAC completion and evidence coverage. Everyday confirmed corrections also inform word practice. Unreviewed or missing transcripts do not become accuracy scores; AAC participation remains separate from speech scoring. Sample counts and missing-data coverage accompany comparisons.

## Clinician dashboard and evidence review

A PIN-gated overview, plan editor and oldest-first review queue provide transcript/evidence filters and direct attempt navigation. Clinicians can inspect consented audio/video, enter notes and reviewer observations, review weekly trends and compare compatible tasks using medians, interquartile ranges, sample counts and descriptive intervals. Communication reports show understanding success among checked exchanges, feedback coverage, taps and seconds from composition to speech start. Playback is not treated as understanding, and response timing is not an acoustic speech-onset measurement.

## Reports and data portability

Printable progress reports, CSV and JSON exports include reporting scope, dates, participant codes, provenance and evidence references, with transcript redaction by default. Recommended passphrase-encrypted JSON reports support decrypt-preview and import; up to ten imported patient snapshots remain read-only and separate from the active profile. Clips require separate consented export and are not embedded in encrypted reports. Personal backups offer encryption, preview and non-overwriting merge, excluding credentials and rehabilitation media. Individual records, recordings and linked reviews can be deleted, or device data erased.

## Privacy controls and local operation

Cloud AI is blocked by default. Device local-only controls, separate context/recent-turn permissions and cross-tab revocation reduce unintended sharing. Approved provider requests use relevant context only; API keys stay in server memory with forget controls and expiry after restart or inactivity. Settings expose provider/model/timeouts and synthetic connection tests. Recognized health/help requests use prepared wording or clarification before models. The installable PWA provides offline core tools, saved content and readiness feedback. Active browser data remains unencrypted; exported reports and backups have separate encryption.

## Navigation and demonstration

Six main tabs organize My space, My phrases, Recent words, My tools, Rehabilitation and Clinician. The root address opens patient Home; older practice/therapist links remain usable. A bilingual 16-token picture-board baseline measures taps and time with Undo, charts and CSV. Seven guarded demonstration scenes, explicitly labelled sample/mock or cached output, silent warm-up and measured stage counters support rehearsals.

**Current boundaries** This is a local prototype, not a clinically validated system or authenticated clinical portal. No HIPAA compliance or CDSCO approval is claimed. Text matching is not acoustic accuracy; personalization does not fine-tune speech models. Generative voice cloning, direct cloud STT/paid TTS, automatic therapist delivery and hosting remain future work.